

UNIT 1: Fundamentals of AI and Data Analytics (30 Marks)

Q1. Differentiate between the following (2 marks each) (6 Marks)

- a) Quantitative data vs Qualitative data
- b) Discrete data vs Continuous data
- c) Nominal data vs Ordinal data

Q2. Star rating for a product is which type of data? (1 Mark)

- a) Discrete b) Continuous c) Nominal d) Ordinal

Q3. Result of rolling a dice is which type of data? (1 Mark)

- a) Discrete b) Continuous c) Nominal d) Ordinal

Q4. Temperature of a city for a past month is which type of data? (1 Mark)

- a) Discrete b) Continuous c) Nominal d) Ordinal

Q5. Course selection from BBA, BCA, BCom is which type of data? (1 Mark)

- a) Discrete b) Continuous c) Nominal d) Ordinal

Q6. EDA stands for? (1 Mark)

- a) Expert Data Analysis
- b) Exploratory Decision Analyst
- c) Exploratory Data Analysis
- d) Expert Document Analysis

Q7. What is EDA? List its objectives. (2 Marks)

Q8. Explain types of EDA. (4 Marks)

Q9. How to identify missing data using Python? (2 Marks)

Q10. How can we handle missing data? (4 Marks)

Q11. What are outliers? How to identify them? Explain with example. (2 Marks)

Q12. What is Gaussian distribution (normal/bell curve) of data? (2 Marks)

Q13. What are the features of normally distributed data? (2 Marks)

Q14. Explain skewness with examples. (4 Marks)

Q15. What is kurtosis? (2 Marks)

UNIT 2: Automate EDA (30 Marks)

Q1. What does automating EDA mean? What are the advantages? (2 Marks)

Q2. Which Python libraries are used to automate EDA? (1 Mark)

Q3. Data structure ndarray is provided by which Python library? (1 Mark)

a) matplotlib b) seaborn c) pandas d) numpy

Q4. Which of the following statistical functions are provided by NumPy? (1 Mark)

a) mean b) mode c) standard deviation d) All of these

Q5. Which Python library allows reading data from CSV, Excel and SQL? (1 Mark)

a) matplotlib b) seaborn c) pandas d) numpy

Q6. How to find total missing values using Python? (1 Mark)

Q7. Which pandas function provides data types and memory details? (1 Mark)

a) df.head() b) df.info() c) df.describe() d) df.isnull()

Q8. What is regression? Explain its features and types. (4 Marks)

Q9. Explain dependent and independent variables using an example. (2 Marks)

Q10. Which of the following is a type of regression? (1 Mark)

a) Linear b) Polynomial c) Logistic d) All of these

Q11. Explain covariance. (2 Marks)

Q12. Explain correlation. (2 Marks)

Q13. What is Machine Learning (ML)? Explain its concepts and benefits. (4 Marks)

Q14. Write a note on ML life cycle. (4 Marks)

Q15. Write a note on supervised and unsupervised learning. (4 Marks)

Q16. Write a note on applications of ML in real world. (4 Marks)

UNIT 3: Understanding Supervised Learning (30 Marks)

Answer the following in short (1 marks each)

1. Define **supervised learning with example**.
2. Define unsupervised learning with example.
3. What is a **label/target** in a dataset?
4. What is meant by a **feature**?
5. Define **validation data**.
6. What is a **loss function**?
7. What is **training data**?
8. What is **testing data**?
9. Explain outliers.
10. Identify the problem is of Classification or Regression
 1. Predict whether an email is **Spam or Not Spam**.
 2. Predict the **price of a house** in rupees.
 3. Predict whether a student will **Pass or Fail**.
 4. Predict the **marks** of a student out of 100.
 5. Predict whether a patient has **Diabetes (Yes/No)**.
 6. Predict the **age** of a person from height and weight.

Answer in Detail (5 marks each)

1. **MSE and MAE Calculation**

A model predicts house prices (in lakhs).

Record	Actual (y)	Predicted ($\hat{y}$)
1	50	52
2	60	59
3	70	68
4	80	78
5	90	92

Tasks :

- a) Compute **MSE**.
 - b) Compute **MAE**.
 - c) Write **one property** of MSE and **one property** of MAE.
2. Differentiate between **classification and regression**.
 3. Explain the concept of **overfitting and underfitting** with example.
 4. Explain **R^2 (coefficient of determination)** with formula and interpretation. Also write **any two limitations** of R^2 .

CHAPTER 4 Vedic Maths 30 marks

- Q1. What is Vedic Mathematics? (2)
- Q2. What is a “Sutra” in Vedic Mathematics? (2)
- Q3. Write the meaning of **Nikhilam Navatashcaramam Dashatah**. (2)
- Q4. Write the meaning of **Ekadhikena Purvena**. (2)
- Q5. Write the meaning of **Paraavartya**. (2)
- Q6. Write the meaning of **Anurupyena**. (2)
- Q7. Write the meaning of **Sankalana-Vyavakalanabhyam**. (2)
- Q8. Write the meaning of **Yavadunam**. (2)
- Q9. Write the meaning of **Vistaran**. (2)
- Q10. Write the meaning of **Chidana**. (2)
- Q11. Write any **two benefits** of using Vedic Mathematics. (2)
- Q12. Write any **two areas/applications** where Vedic Mathematics is useful. (2)
- Q13. What does “**Shunyam Saamyasamuccaye**” mean? (2)
- Q14. What does “**Sesanyan**” mean? (2)
- Q15. What does “**Rupan**” mean? (2)